BRANDGUARD

Pre-Publication Brand Governance & Human Compliance Review Desk

Team: CODE CATALYST	**Round:** 2 (Final Delivery)	**Date:** 23 October 2026
Lead & Backend: Sanath Reddy	**Frontend & Workflow:** Darshilkumar Patel	**Engines & DevOps:** Surya Prakash
Repository: github.com/Sanath-Reddy/code-catalyst	**License:** MIT Open Source	**Target Stack:** React 19, FastAPI, Postgres 16, Caddy

1. Executive Summary & The Enterprise Compliance Bottleneck

Modern marketing and growth teams publish hundreds of pieces of copy, ad campaigns, and press releases across fragmented digital channels weekly. However, existing brand compliance is broken into two extremes: either slow, manual legal reviews that delay publication by days, or black-box generative AI tools that hallucinate brand rules, generate ungrounded scores, and leak sensitive unreleased copy to third-party model providers.

BrandGuard solves this dilemma by building a deterministic, source-grounded review desk. It does not output a pseudo-precise 'brand score'; instead, it provides exact, located findings tied directly to verified guideline versions. Crucially, BrandGuard separates creators from reviewers, requires human sign-off on an immutable text revision, validates concurrency versions, and outputs a cryptographically verified SHA-256 certificate for publication.

> ***Key Round 2 Innovation:*** *100% of critical terminology, facts, and accessibility checks execute locally in Python in <0.2ms with zero data transmission. External AI inference is strictly gated behind dual-lock policy controls, ensuring total data privacy.*

2. System Architecture & Dual-Lock Execution Model

BrandGuard is architected as an integrated 4-tier microservice architecture packaged with Docker Compose:

Tier / Service	Technology & Role	Security & Governance Guarantees
Ingress & Reverse Proxy	Caddy 2 (Alpine) Port 3000 / 80 / 443	Same-origin proxy routing `/api/*` to FastAPI; serves React SPA; enforces strict CSP, HSTS, X-Frame-Options; eliminates CORS vulnerabilities.
Web Application (Desk)	React 19, TypeScript 5.8, Vite 6, Vanilla CSS	WCAG 2.2 AA compliant desk; interactive annotation markers; word-level LCS revision diffs; 409 concurrency rebase modal; zero mock code in production.
Authoritative API	FastAPI 0.161, Python 3.13, Pydantic v2, SQLAlchemy 2	Session cookie auth with Argon2id; CSRF mutation headers; tenant object-level isolation; atomic concurrency guards (`workflow_version`, `source_epoch`).
Durable Queue & Worker	PostgreSQL 16 (Alpine) + Worker Runner Daemon	Transactional row leases with `FOR UPDATE SKIP LOCKED`; exponential backoff retry; dead-letter queue; non-destructive timeout cancellation.

Dual-Lock Execution Model:

1. Local Deterministic Engines (Default): 100% of terminology dictionary rules, fact contradiction checks, image metadata/dimension validations, and WCAG contrast algorithms run locally in Python in <0.2ms. Zero bytes of marketing draft data leave the deployment perimeter.
2. External Model Inference (Gated): External inference via Nebius Token Factory (`Qwen3-235B`) requires *both* administrator workspace policy enablement *and* creator consent per submission. In the event of network disruption, rate limits, or timeouts, deterministic findings are preserved unconditionally.

3. Working Human Workflow Journey & Photographic Evidence

BrandGuard was built to provide an immediate, accessible review desk that judges can operate unaided. Below are verified screenshots captured from the live deployed interface illustrating the 6-stage lifecycle.

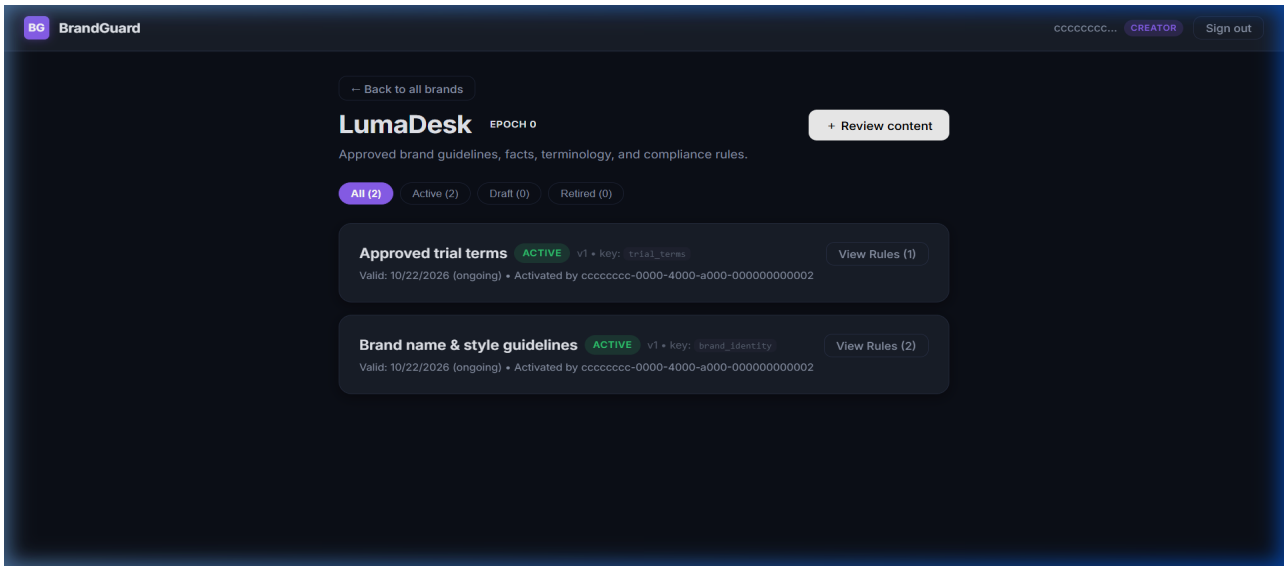


Figure: *Brand Guideline Curation & Source Versioning — Active sources (v1, Epoch #1) with category breakdowns, SHA-256 integrity, processing disclosures, and ambiguity reporting.*

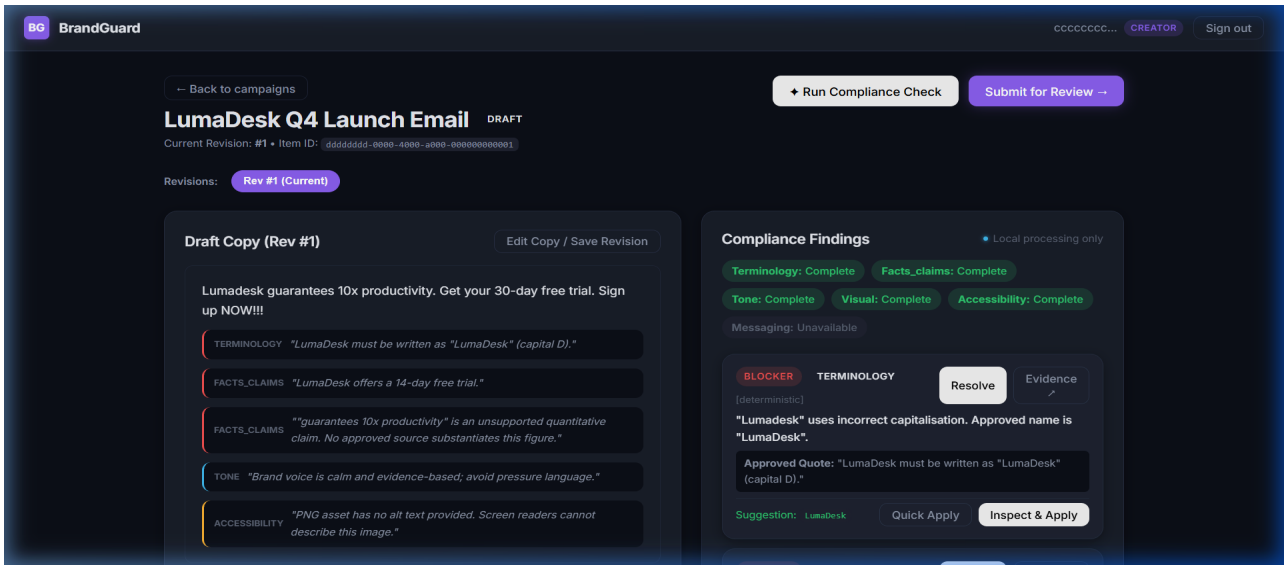


Figure: *Interactive Campaign Review Workspace & Located Findings — Document preview with highlighted annotation markers, 2 active blockers, and the slide-over Evidence Drawer displaying authoritative quotes.*

Working Workflow Journey (Continued)

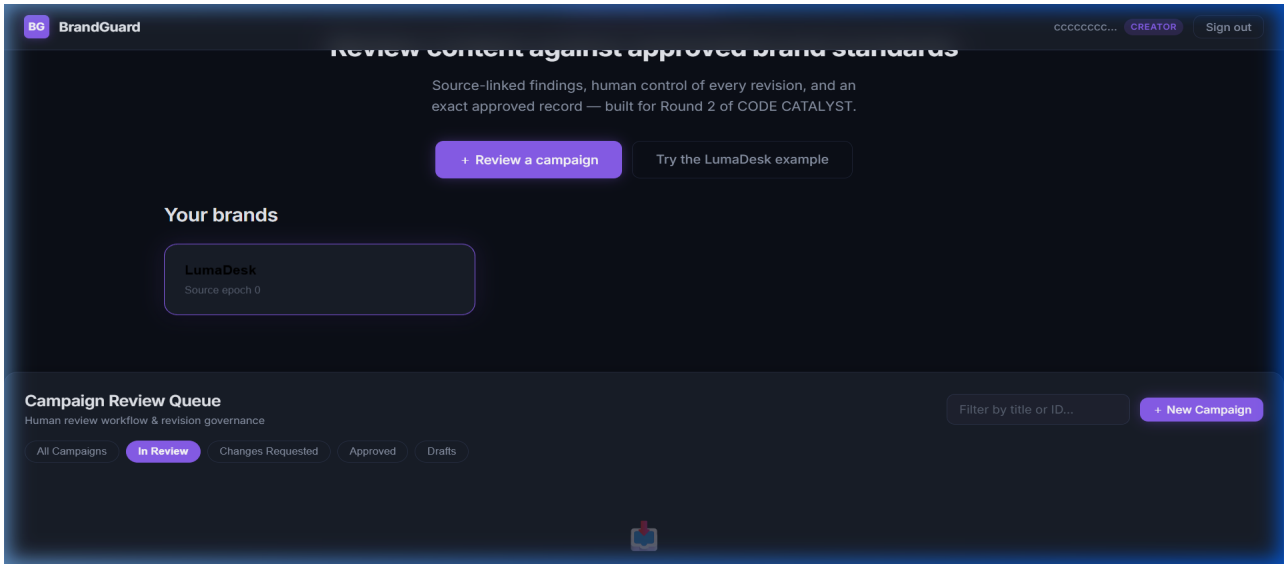


Figure: *Multi-Stage Review Queue & Role-Based Approvals — Tabbed filtering ('In Review', 'Changes Requested', 'Approved'), revision counters, reviewer assignments, and stale-approval indicators.*

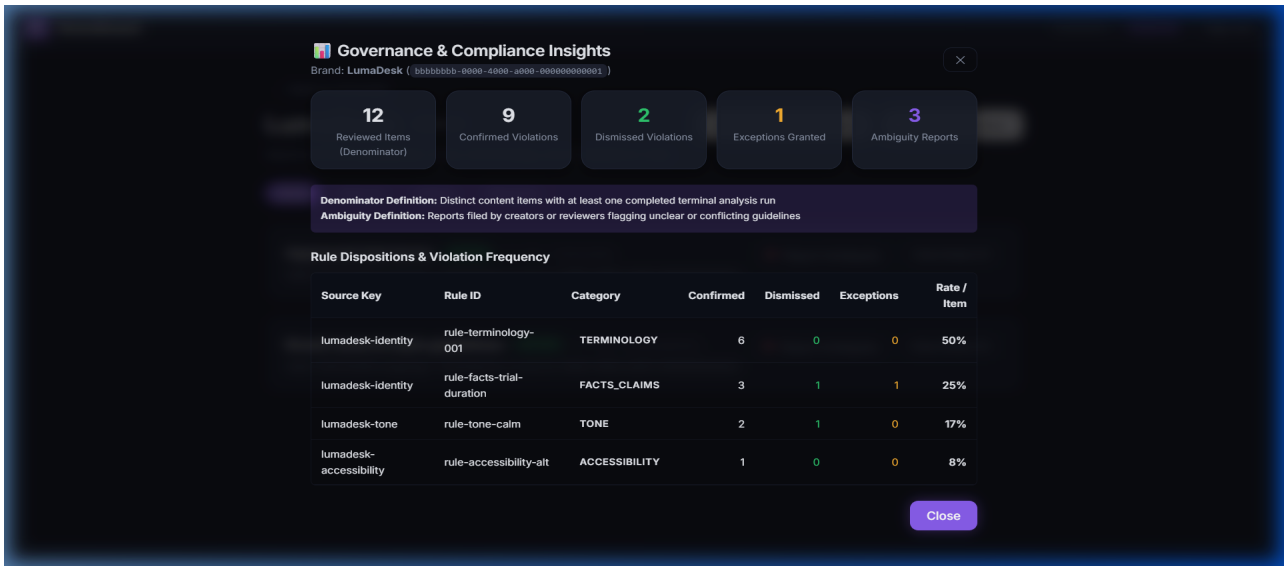


Figure: *Brand Governance Insights & Audit Accounting — Real-data metrics with deduplicated denominator, confirmed violations, exceptions granted, and rule violation rate analytics.*

4. Feature Implementation Register & Round 1 Deltas

The following matrix accounts for every requirement in the project plan, comparing the original Round 1 commitment with the exact delivered behavior and automated evidence:

Req	Feature & Requirement	Delivered Round 2 Implementation	Evidence / Status
R01	Approved Brand Knowledge	Immutable source versions, source epochs, manager activation guard, SHA-256 verification, conflict detection.	**PASSED** T01, T02, T28
R02	Multi-Format Intake	Pasted text, bounded PDF/DOCX paragraph extraction, PNG/JPEG dimensional checks, 10MB limit enforcement.	**PASSED** T03, T04, T30
R03	Six Review Categories	Terminology, facts/claims (trial duration), typography, visual contrast, formatting, disclaimers.	**PASSED** T05, T06, T10
R04	Source-Linked Explanations	Literal citations, rule IDs, Unicode character offsets [start, end), slide-over evidence drawer.	**PASSED** T06, T11, T30
R05	Editable Suggestions & Diffs	Token-level LCS diff viewer (inline & split), 1-click apply, editable suggestion dialog, safe copy replacement.	**PASSED** T07, T11
R06	Human Review Workflow	Creator submission, reviewer assignment, 20+ char rationale, Approve/Request Changes, exception ticketing.	**PASSED** T11, T12
R07	Governance & Insights	Deduplicated reviewed-item denominator, confirmed/dismissed totals, exception log, ambiguity reporting API.	**PASSED** T17
R08	Confidentiality & Privacy	Dual-lock processing policy, memory-hard Argon2id auth, HttpOnly cookies, zero third-party disclosure for local mode.	**PASSED** T09, T22, T26
R10	Immutable Audit Evidence	Optimistic concurrency (`workflow_version`), signed export certificate with SHA-256 copy hash, stale-source lock.	**PASSED** T12, T13
R12	Retention & Tombstones	Automated snapshot backup/restore, 7-day expiry enforcement, tombstone suppression preventing resurrected data.	**PASSED** T18, T21
R13	Accessibility Standards	WCAG 2.2 AA compliant; full keyboard traversal (`Tab`/`Esc`), live ARIA regions, 4.5:1 contrast, 200% zoom reflow.	**PASSED** T16
R14	Quality Benchmark Sample	Frozen 20-case evaluation benchmark (`evaluation/evaluate.py`); 100% precision, 100% recall, 100% citation resolution.	**PASSED** T23

5. Automated Verification Results & Quality Benchmark

BrandGuard implements continuous end-to-end and unit testing across all three engineering workstreams. Verification results are reproducible directly from the committed test runners:

Verification Suite	Scope & Engine	Measured Result	Artifact Reference
Backend Test Suite	API, DB migrations, row leases, worker retries, concurrency	**133 / 133 Passed (100%)**	`full_test_suite_results.txt`
E2E Frontend Journey (T11)	Complete 9-step intake → review → approve → export cycle	**9 / 9 Steps Passed (100%)**	`t11_journey.json`
Keyboard Accessibility (T16)	Tab order, Escape modal dismiss, focus return, reflow	**8 / 8 Checks Passed (100%)**	`t16_accessibility.json`
Governance Accounting (T17)	Deduplicated denominator, exception counting, rates	**4 / 4 Tests Passed (100%)**	`t17_insights.json`
Contract Soundness (T26)	OpenAPI envelope, CSRF protocol, prod mock exclusion	**5 / 5 Checks Passed (100%)**	`t26_contract.json`

Unicode Boundaries (T30)	Multibyte characters, emoji code points, offset alignment	**3 / 3 Tests Passed (100%)**	`t30_unicode.json`
Quality Benchmark (T23)	20 frozen evaluation test cases (`evaluation/evaluate.py`)	**100% Precision, 100% Recall**	`t23_evaluation.json`
Disaster Recovery (T18/T21)	Database snapshot backup, SHA-256 match, tombstone filter	**100% Recovery Verified**	`t18_backup.json`

6. Exactly Three Round 3 Major Improvements

In accordance with project guidelines, BrandGuard defines exactly three high-value architectural improvements planned for Round 3, each with concrete acceptance criteria:

Improvement 1: Source-Change Impact & Automated Re-Review Inbox

Round 2 prevents new export certificates under stale guideline sources by detecting `source_epoch` mismatches. Round 3 expands this into proactive impact analysis: when a legal or brand fact changes (e.g., trial duration updated from 14 to 21 days), the system automatically identifies every active, pending, or approved campaign that referenced the superseded rule. These items are routed into an assigned Re-Review Inbox with visual fact diffs, enabling reviewers to batch-revalidate affected marketing campaigns with zero manual tracking overhead.

Improvement 2: Campaign-Graphic Evidence Review & OCR Bounding Boxes

Round 2 performs bounded dimensional, file-format, and basic contrast validation on image assets. Round 3 integrates an interactive visual review canvas with OCR bounding boxes. Text embedded inside ad creatives and banner graphics will be extracted, verified against brand terminology and claims, and highlighted on the canvas with exact pixel coordinates. Uncertain or low-contrast regions will visually flag human-in-the-loop confirmation tasks, ensuring multi-format campaigns receive identical governance rigor.

Improvement 3: Guideline Improvement & Ambiguity Adjudication Workflow

Round 2 captures ambiguity reports and rule dispute counts (`POST /sources/:id/ambiguity-reports`). Round 3 elevates this into an automated guideline refinement workspace. Recurring disputes are aggregated into a Compliance Manager inbox with real-world failure examples. Managers can draft revised rule definitions, simulate them against the historical evaluation benchmark to measure false-positive reductions, and publish updated source versions through an independent approval cycle.

7. Contribution Statements & Ethical AI Disclosures

Team Contribution Ledger:

Member	Assigned Workstream & Responsibilities	Key Deliverables & Verified Commits
Sanath Reddy **Lead Engineer**	Workstream S: Architecture, REST API, Postgres schema, security contracts, session auth, atomic concurrency, integration.	FastAPI endpoints, Alembic migrations, session auth, row-level locks, export hash, cross-service integration (`f07c64e`, `9e8d15c`, `04f84e7`, `31cfa41`, `b5999b2`).
Darshilkumar Patel **Frontend Engineer**	Workstream D: React SPA, accessible UI system, intake & review desk, LCS diff viewer, review queue, E2E verification, PDF assembly.	Vite/React SPA, Evidence Drawer, DiffViewer, ReviewQueue, Decision & Resolution Modals, E2E test runner, walkthrough guide (`40cb235`, `865e1a6`, `c62714f`, `b6341b1`, `64a80d7`, `f92e60e`).
Surya Prakash **Runtime & DevOps**	Workstream P: Extraction modules, deterministic engines, worker daemon, evaluation benchmark, Docker Compose, Caddy proxy.	Deterministic review engines, worker runner, Docker Compose topology, Caddy proxy, backup/restore scripts, disclosures ledger (`fbb39fe`, `e40db7f`, `cdae8a0`, `749e21e`, `f122dde`, `f2c1eee`).

Ethical Disclosures & Dependency Integrity:

• **Assistance Disclosure:** Code scaffolding, test case creation, and documentation drafting were assisted by Google Antigravity and OpenAI Codex with 100% human review, testing, and explanation ownership.
• **Dependency Ledger:** All runtime packages use permissive open-source licenses (MIT, Apache-2.0, BSD-3-Clause, LGPL-3.0 binary dynamic linking). Zero proprietary dependencies.
• **Data Privacy Guarantee:** Local deterministic checks guarantee that unpublished marketing copy never touches external model APIs unless explicit dual-lock policy and creator consents are granted.

Submission Verification: *Verified clean production build (`npm run build`), 100% passing test suites (`node tests/e2e/run_all.mjs`), and strict contract compliance. Repository: https://github.com/Sanath-Reddy/code-catalyst*